

Name: Team Reed**Date:** February 5, 2026**Class:** Engineering

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a [decision matrix](#).

Create a plan of work, ideally using a [Gantt chart](#) that may be housed in this document or in a different day-to-day document.

Generate a design blueprint with enough details that someone else not familiar with your project could also build your design. Think of this as your first sketch.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

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Date: February 5, 2026

Class: Engineering

Design Idea #1

Date: February 9, 2026

Description: Gold Pan Shaker

Design Idea #2

Date: February 9, 2026

Description: Build and set up some kind of filtering system

Design Idea #3

Date: February 9, 2026

Description: A mesh net to put into the water so it will collect the microplastics

Design Idea #4

Date: February 9, 2026

Description: An arm to hold out the thing that is holding the net

Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

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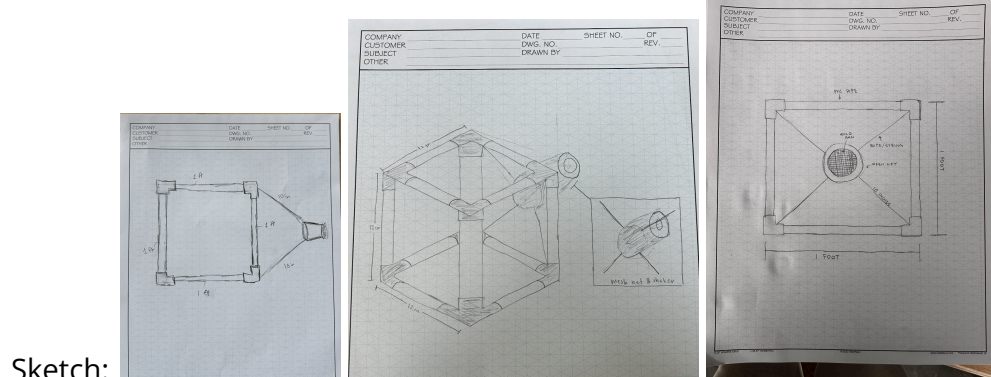
	weight	Filtering System	Gold pan shaker	Meshnet	Arm to hold net
Criterion A	17.8%	1	2	2	.25
Criterion B	21.4%	1	1.25	1.75	.25
Criterion C	10.7%	1.5	1	1.25	.75
Criterion D	14.3%	1.75	.5	1.25	-.25
Criterion E	7.2%	1	1.25	.75	-.75
Criterion F	25%	.25	1	.5	-1
Criterion G	3.6%	4	1	1	1
score	100%	2.6915	3.7224	3.6514	-0.24852

Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date:



Sketch:

Manufacturing Methods:

Step 1: Remove the lip on top of the Gold pan

Step 2: Add the mesh net around the gold paner

Step 3: Attach a string connecting to the mesh net to the ROV

Supply List and Costs:

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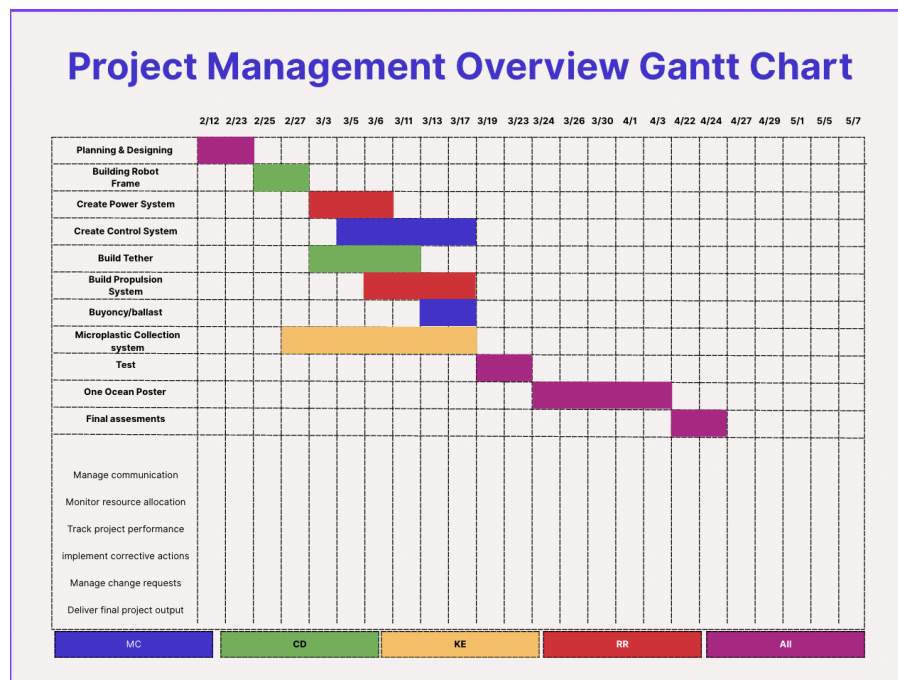
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Item	Cost each	# needed	Total
Gold Panner	\$18.45	1	\$18.45
Mesh Nets	\$9.99	2	\$19.98
Total Cost			

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.



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